# Adhithyaa A

Linkedin: linkedin.com/in/AdhithyaaAnurag
Email: thakkuduunni2005@gmail.com
Github: github.com/AnuragAdhithyaa
Mobile: +91-8220491811

## SKILLS

- **Languages**: C/C++ for Embedded Systems, Python for IOT Application, Basic Verilog for Digital Design
- **IOT Platforms**: Arduino IDE, NodeMCU ESP32 Development, Raspberry Pi
- **Embedded Systems**: Sensor Interfacing (I2C,SPI,UART), Wi-FI, MQTT
- **Digital VLSI Design**:Xilinx Vivado
- **Soft Skills**: Problem Solving, Public Speaking, Team Player, Project Management, Adaptability

## INTERNSHIP

**AAI – Airport Authority of India** June 2025 – July 2025
*Student Intern*

- **About**: Assisted in understanding and monitoring communication systems (HF, VHF, CPDLC). Gained exposure to navigation systems such as Localizer, Glide Path and DME integration. Learned about surveillance technologies including PSR, MSSR, ASMGCS and MLAT operations.

- **Tech stacks used**: HF & VHF radio communication systems, Instrument Landing System (ILS) – Localizer/Glide Path/DME, CPDLC, Primary & Secondary Surveillance Radar (PSR/MSSR), ASMGCS, MLAT, signal filters & rectifiers.

**Automate X – PLC based training to Industrial Automation** June 2025 – July 2025
*Student Intern*

- **About**: Enhanced knowledge in PLC and Ladder Logic through designing and troubleshooting industrial control systems using the LogixPro and Twidosuite simulation software. Created and simulated the complete control logic for a PLC-based conveyor belt simulator in Twidosuite, accurately modelling the machine's full operational sequence. Applied automation principles on batch simulation and automatic opening and closing of valves with accurate timing of the counters.

- **Tech used**: LogixPro, Twidosuite

## PROJECTS

**Industrial Document Analyzer (AI Powered)**: (**GitHub**) August 2025- September 2025

- Designed and implemented an automated document analysis system for industrial records.
- Built a pipeline to scan, extract and classify data from invoices, maintenance logs, and compliance documents.
- Integrated with a simple dashboard / database to store and retrieve processed documents.

**Tech**: Python, Hugging Face Transformers, pdfplumber, YAKE

**Enviro-Desk – Portable environmental desk monitoring setup** (**GitHub**) April 2025 – May 2025

- Developed an advanced environmental monitoring system using Raspberry Pi 4, integrating DHT22, MQ135, and soil moisture sensors to deliver real-time data via a professional web dashboard.
- Led end-to-end project execution, from hardware setup and sensor calibration to Flask-based backend development and responsive frontend design.
- Demonstrated expertise in IoT, Python, and web technologies, ensuring robust data acquisition and visualization for impactful insights.

**Tech:** Fusion 360, Arduino IDE, Blynk-IOT

**PLC based Conveyor Belt Automation System**: June 2025 – July 2025

- Designed and implemented a PLC-controlled conveyor belt system to automate material movement.
- Developed and tested ladder logic to control start/stop sequences, speed, and direction.
- Verified safety interlocks and optimized cycle times for efficient throughput.

**Tech**: Twidosuite Siemens/Allen-Bradley PLC (model), Ladder Logic Programming

**32-Bit RISC Processor**: April 2025 – May 2025

- Designed and implemented a 32-bit RISC processor architecture entirely in Verilog HDL using Xilinx Vivado.
- Developed core modules including a pipelined datapath, program counter with branch/stall handling, and instruction decode logic with decoder and register file.
- Simulated and verified functionality in Vivado using testbenches; no hardware implementation was required.
- Documented architecture, control logic, and simulation results for future FPGA deployment.

**Tech:** Verilog HDL, Xilinx Vivado, Testbenches & Simulation, Digital Design Principles

## CERTIFICATES

- Deloitte Certification on Data Analyst Job Simulation July 2025-September 2025
- Automate X Mastering PLC for Industrial Automation June 2025-July 2025

## ACHIEVEMENTS

**Tech- Spark Technical Quiz Competition**: October 2024

- 3rd Runner up among more than 50 teams in technical Quiz Competition held by the University

**Analytical Leaderboard**: November2025-December 2025

- Only Student representing School of Electronics among the top 60 analytical & reasoning leaderboard conducted all over the schools in the university

## POSITIONS AND RESPONSIBILITY

**Student Academic Advisory Committee Member at SEEE** August 2025-present
Lovely Professional University

**Vice President** September2023-August2025
Silicon Forest Club, LPU

**Smart India Hackathon Coordinator** September2025
Lovely Professional University

## EDUCATION

**Lovely Professional University** Punjab, India
*Bachelor of Technology – Electronics and Electric Engineering- Internet of Things;* ***CGPA: 7.8*** *August 2023- Present*

**Orchard Schools** Tiruchirappalli, Tamil Nadu
*Intermediate;* ***Percentage: 80%*** *April 2020 - March 2022*

**Orchard Schools** Tiruchirappalli, Tamil Nadu
*Matriculation;* ***Percentage: All Pass 100%*** *April 2019 - March 2020*